



Problem Statement

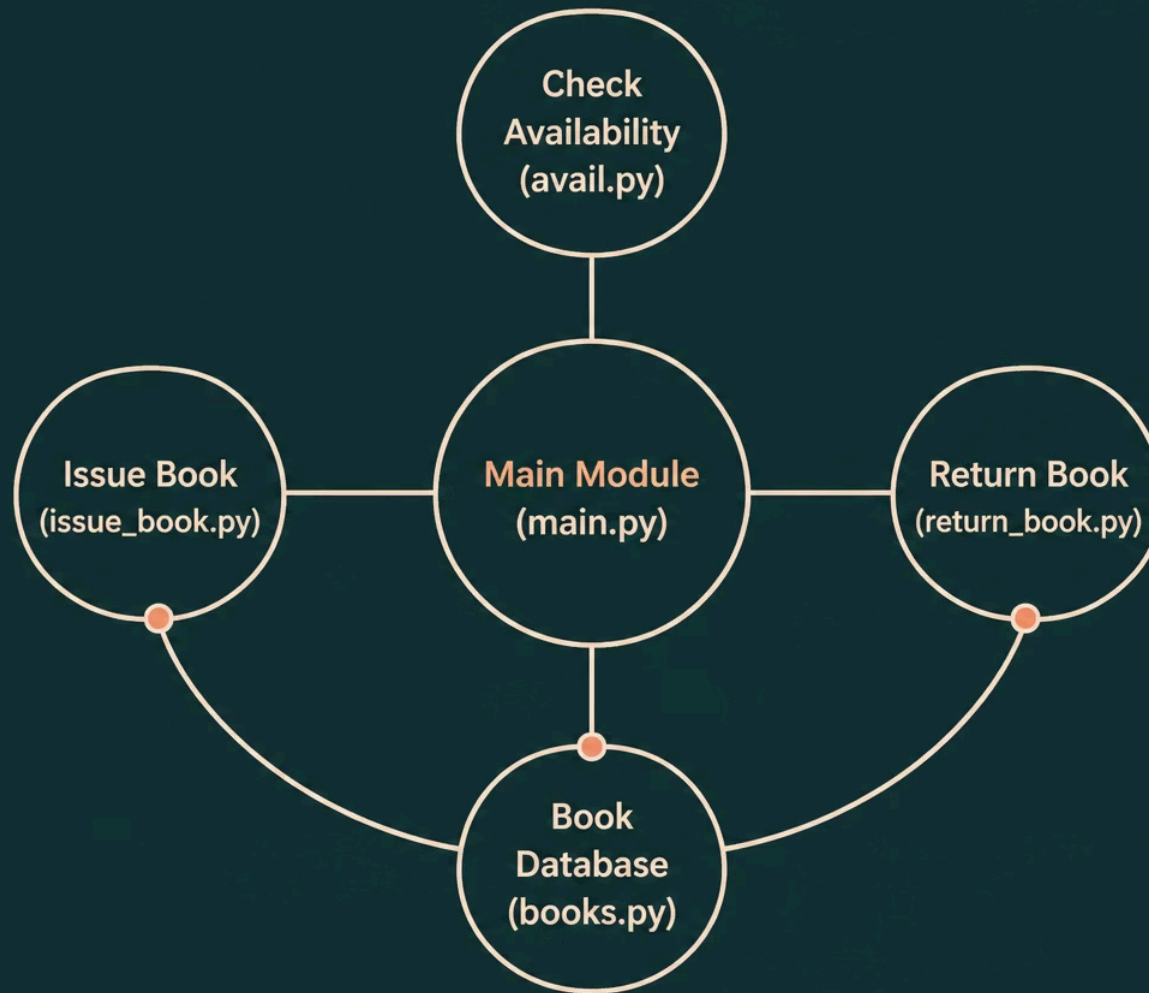
- Traditional library management still relies on manual record-keeping, slowing down daily operations.
- Tracking book availability, issues, and returns is difficult and often inconsistent.
- These manual workflows lead to frequent errors, delays, and reduced efficiency.
- A better system is needed to improve accuracy, speed, and overall library performance.



Proposed Solution

The Digital Library System streamlines library operations by digitizing book availability, issue, and return tracking. It automatically updates copies, reduces manual work, improves accuracy, and speeds up book management.

System Architecture



Module 1 – Book Database (books.py)

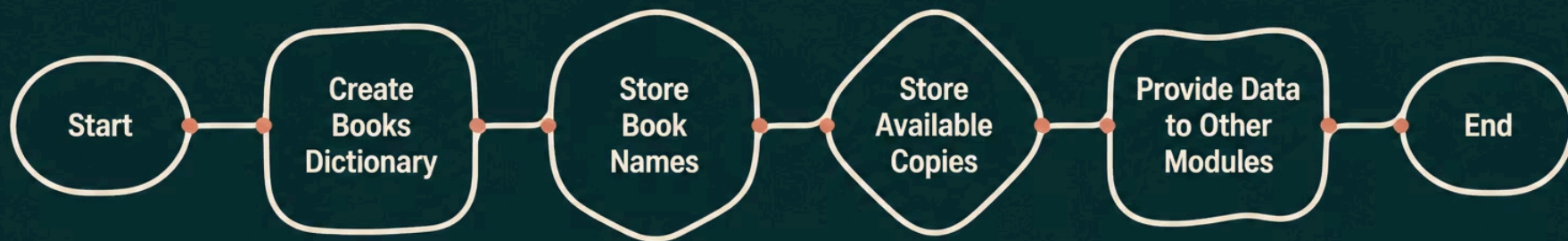
Purpose

Stores book records and available copies as the central database for the system.

Explanation

Initializes the library data, tracks books in a dictionary, and shares it with other modules.

Flow



Module 2 – Check Availability (avail.py)

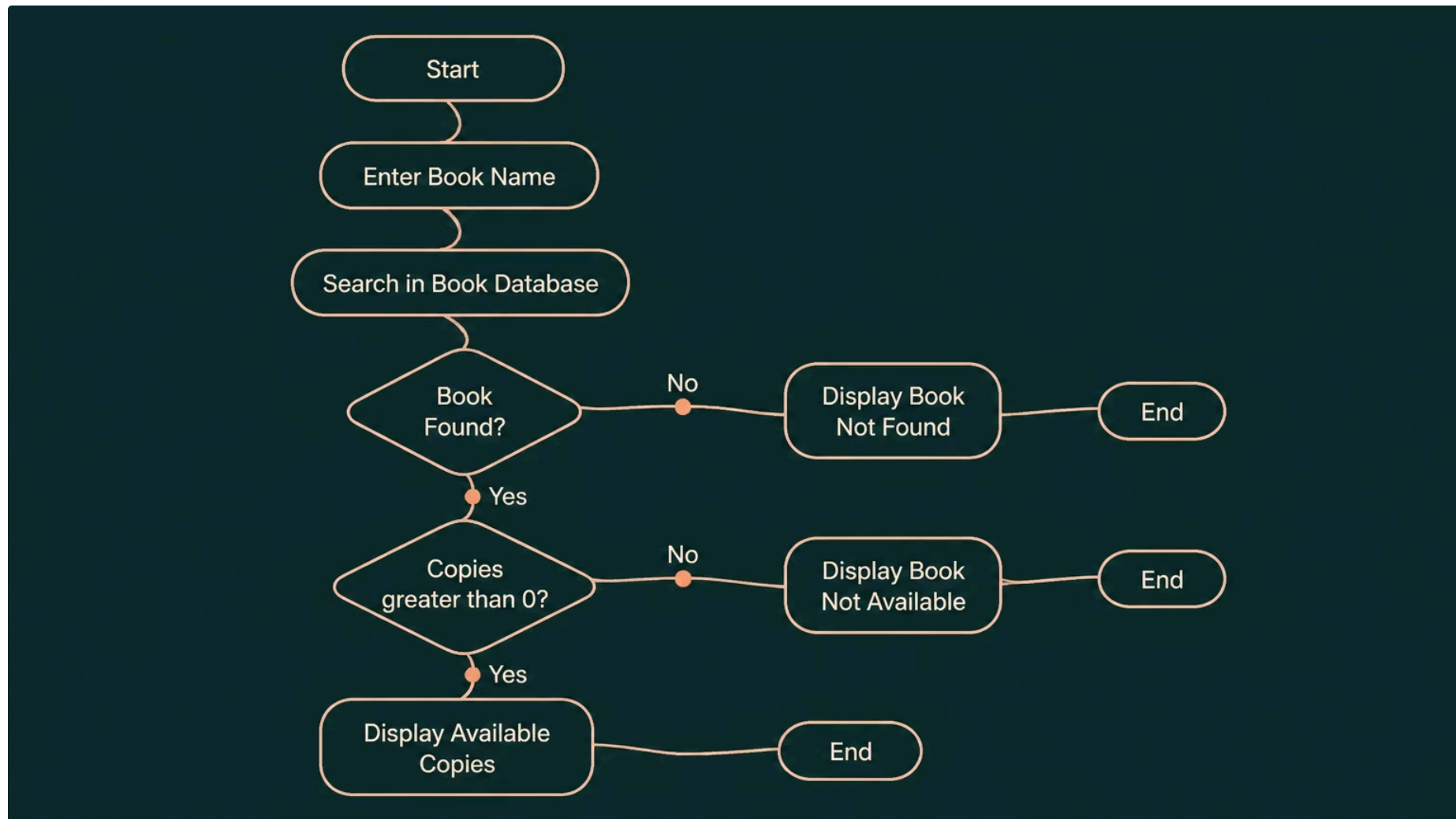
Purpose

Checks whether a book is available.

Explanation

Enter a book name, search the database, and show availability and copy count.

Flow



Module 3 – Issue Book (issue_book.py)

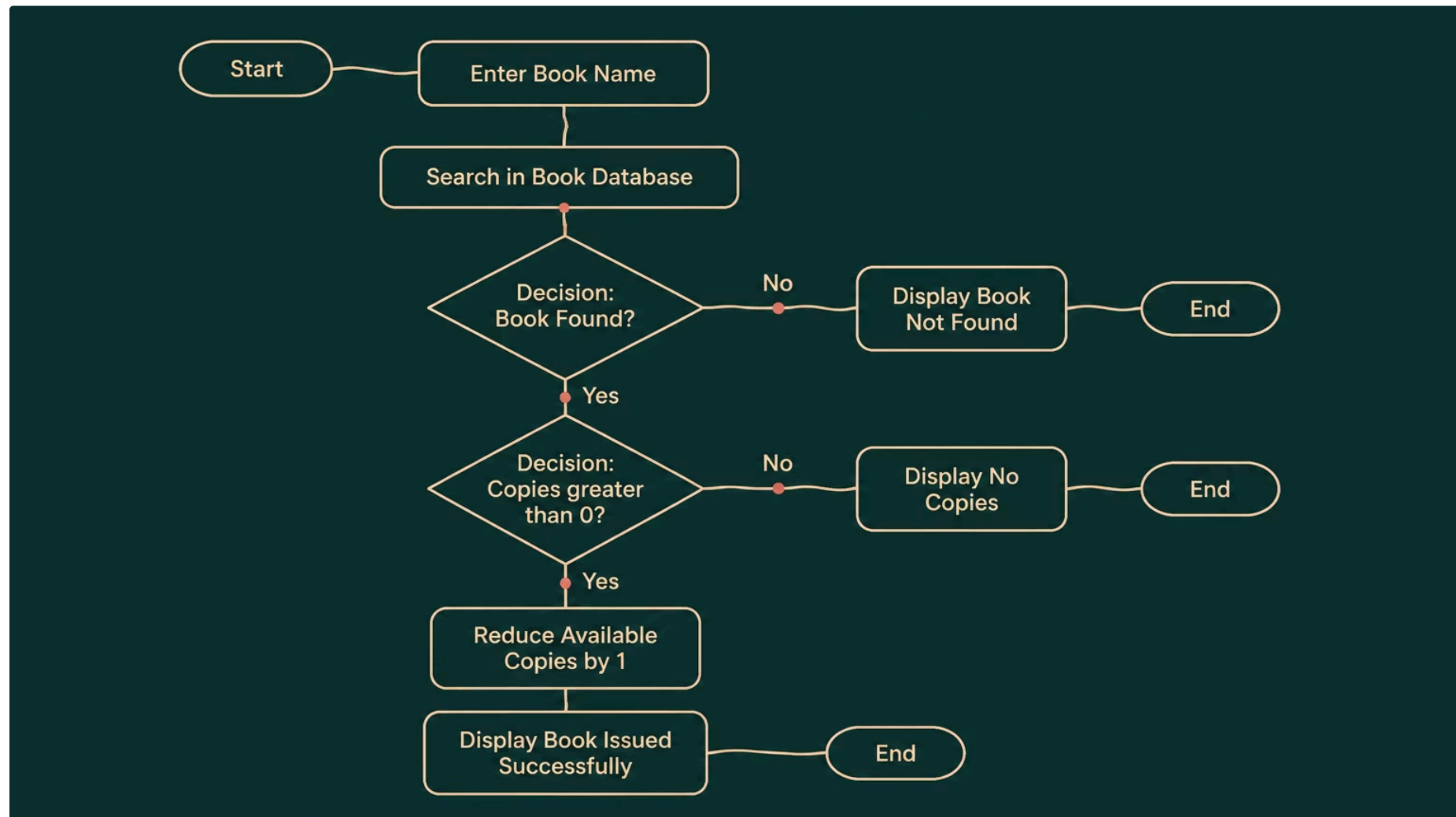
Purpose

Issues a book and updates the available copies.

Explanation

- Gets the requested book name
- Checks if the book exists
- Confirms copies are available
- Reduces the copy count after issuing
- Shows a confirmation message

Flow



Module 4 – Return Book (return_book.py)

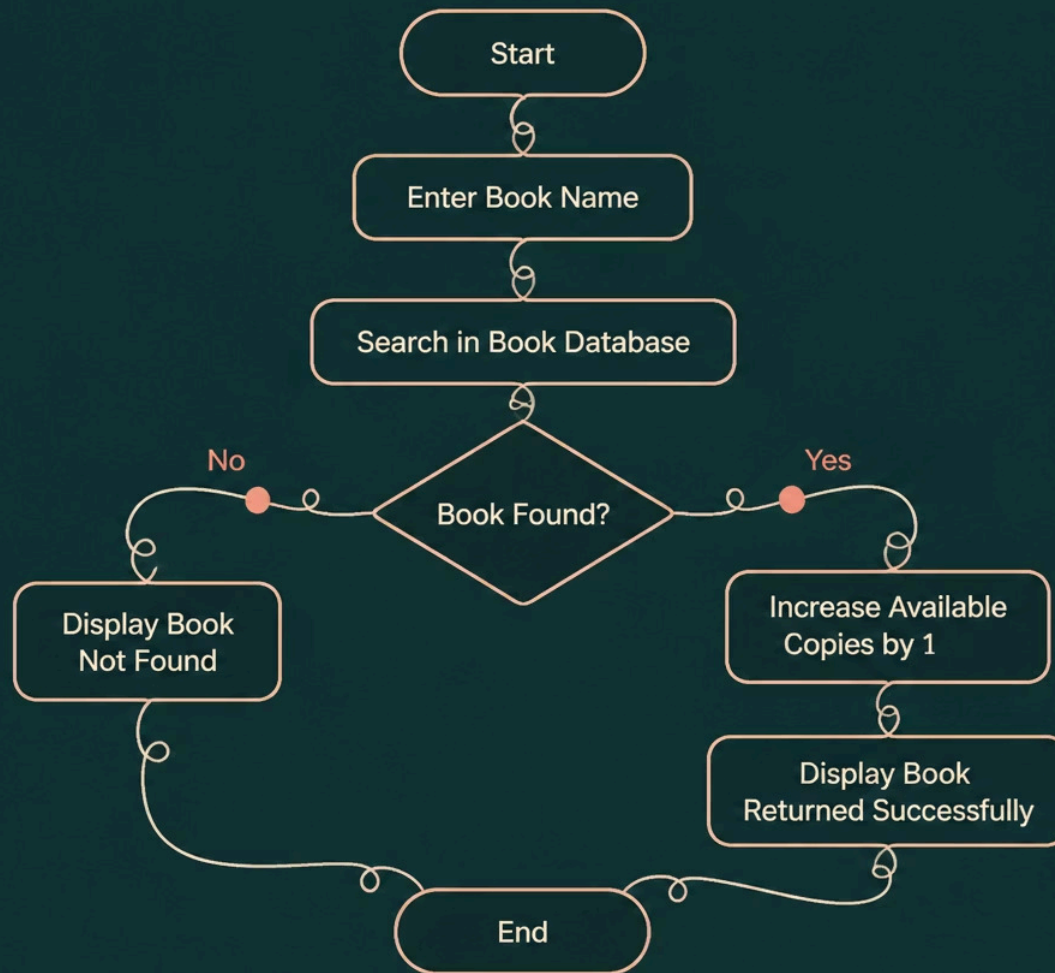
Purpose

Updates available copies after a book is returned.

Explanation

- Receives the returned book name
- Checks whether the book exists
- Increases the copy count
- Shows a success message

Flow



Module 5 – Main Module (main.py)

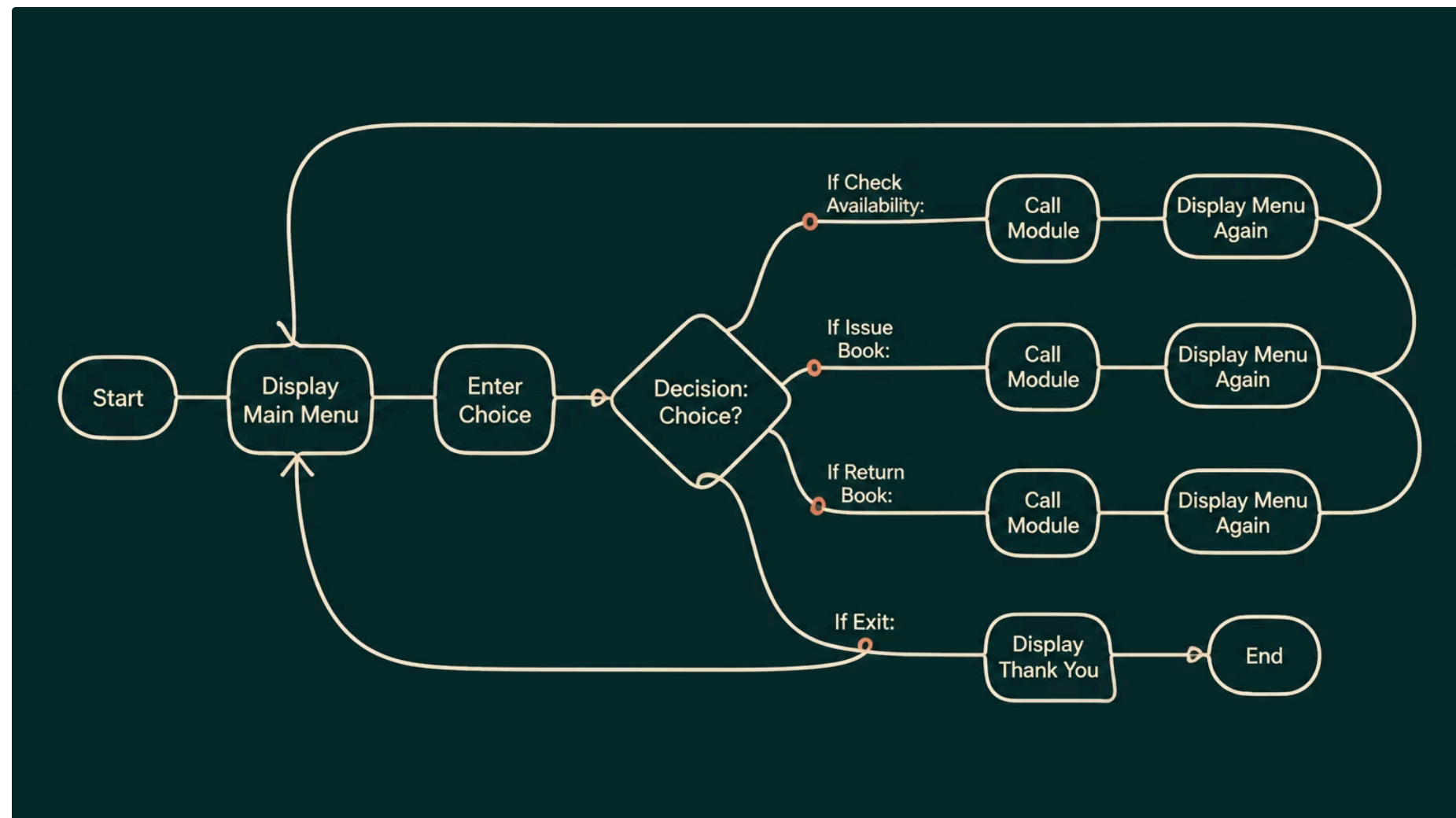
Purpose

Manages the Digital Library System and connects all modules.

Explanation

Shows the main menu, accepts a choice, calls the selected module, and repeats until Exit.

Flow



Technology Used

Technology	Purpose
Python	Core language used to build the system.
Python Functions	Handle tasks like checking availability, issuing books, and returning books.
Python Modules	Split the app into separate files for easier management.
Dictionary (Data Structure)	Stores book names and copy counts using key-value pairs.
Conditional Statements (if, elif, else)	Support decisions based on user input and availability.
Loops (while)	Repeat the main menu until exit.
Input/Output Functions	Take input and display results in the console.

Future Implementation

- Use a real database like MySQL or SQLite for permanent storage.
- Add login and authentication for librarians and students.
- Track member records and borrowing history.
- Enable search by title, author, or category.
- Add due dates and overdue fine calculation.
- Generate reports on issued, returned, and available books.
- Build a web or mobile app for remote access.
- Use barcode or QR scanning for faster issue and return.
- Send email or SMS reminders for due dates and overdue books.

Thank You

Thank you for your time!

We hope the Digital Library System shows how digital technology can simplify library operations.

Presented By

Team Member 1 • Team Member 2 • Team Member 3 • Team Member 4